

Method-level Summary — Models

Setup. To compare methods on equal footing, we restrict the summary to the four models for which every (method × dataset) cell has full data — gpt-4.1, gpt-4o, gpt-5, and gpt-5.1. Each model was evaluated over 10 runs × 100 preference pairs across both airline and retail datasets. For each method we report mean ± std per sample of latency, prompt and completion tokens, and a token decomposition: rollout tokens (the simulator's full extension of the candidate trajectory; 0 for the prompt-critic methods) plus judge tokens (the final verifier scoring call). Their sum is reported as total tokens.

Method	Latency Mean	Latency Std	Prompt Mean	Prompt Std	Completion Mean	Completion Std	Rollout Mean	Rollout Std	Judge Mean	Judge Std	Total Mean	Total Std
ToolRM	11.14	11.68	6,644	903	907	957	0	0	7,552	1,362	7,552	1,362
Analysis-based	13.86	12.64	6,798	903	1,266	1,156	0	0	8,065	1,521	8,065	1,521
Simulation (DB)	142.71	129.03	19,705	8,274	2,186	2,708	12,131	8,286	9,760	2,276	21,891	9,527
Simulation (tools)	78.43	82.60	26,535	5,152	2,866	3,175	18,977	5,010	10,424	2,497	29,400	6,725